

The global martian volcanic evolutionary history

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ARTICLE INFO

Article history:

Received 13 February 2008

Revised 27 November 2008

Accepted 2 December 2008

Available online 31 December 2008

Keywords:

Mars

Impact processes

Mars, surface

ABSTRACT

Viking mission image data revealed the total spatial extent of preserved volcanic surface on Mars. One of the dominating surface expressions is Olympus Mons and the surrounding volcanic province Tharsis. Earlier studies of the global volcanic sequence of events based on stratigraphic relationships and crater count statistics were limited to the image resolution of the Viking orbiter camera. Here, a global investigation based on high-resolution image data gathered by the High-Resolution Stereo Camera (HRSC) during the first years of Mars Express orbiting around Mars is presented. Additionally, Mars Orbiter Camera (MOC) and Thermal Emission Imaging System (THEMIS) images were used for more detailed and complementary information. The results reveal global volcanism during the Noachian period (>3.7 Ga) followed by more focused vent volcanism in three (Tharsis, Elysium, and Circum-Hellas) and later two (Tharsis and Elysium) volcanic provinces. Finally, the volcanic activity became localized to the Tharsis region (about 1.6 Ga ago), where volcanism was active until very recently (200–100 Ma). These age results were expected from radiometric dating of martian meteorites but now verified for extended geological units, mainly found in the Tharsis Montes surroundings, showing prolonged volcanism for more than 3.5 billions years. The volcanic activity on Mars appears episodic, but decaying in intensity and localizing in space. The spatial and temporal extent of martian volcanism based on crater count statistics now provides a much better database for modelling the thermodynamic evolution of Mars.

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1. Introduction

Volcanism at the surface of Mars is extensive but not uniformly distributed (Fig. 1) and includes a diversity of volcanic landforms such as central volcanoes, tholi, paterae and small domes, as well as vast volcanic plains. This is reflected in the comprehensive collection of volcanic features based on Viking images compiled by Hodges and Moore (1994). Such a diversity of landforms implies different eruption styles and probable changes in the style of volcanism with time, as well as interaction with the martian cryosphere and atmosphere during the evolution of Mars. Many volcanic constructs are associated with regional tectonic or local deformational features. Two topographically dominating and morphologically distinct volcanic provinces are the Tharsis and Elysium regions, whose morphologies are strongly analogous to basaltic volcanic landforms on Earth. Both are situated close to the equator on the dichotomy boundary between the cratered (older) highlands and the northern lowlands and are approximately 120° apart. The huge volcanoes in the Tharsis region (Olympus Mons, Ascraeus Mons, Pavonis Mons, and Arsia Mons) share many characteristics with Hawaiian basaltic shield volcanoes (Carr, 1973). They are constructed from multiple lobate flows of lava (Bleacher et al., 2007; Hiesinger et al., 2007), and show complex nested coalescing sum-

mit calderas of varying age, and gentle slopes on the order of a few degrees. Their eruption style appears effusive and relatively non-explosive in nature, although there are indications of composite volcano construction (Head and Wilson, 1998; Head et al., 1998a). The main differences between the martian and terrestrial volcanoes are the greater sizes and lengths of flows of the martian examples, mainly due to higher eruption rates, the “stationary” character of the source (no plate tectonics) and the lower gravity (Basaltic Volcanism Study Project, henceforth BSVP, 1981). Plescia and Saunders (1979) summarized the chronology and classification of martian volcanic activity based on the Viking imagery data, and grouped the volcanic landforms into (1) shield volcanoes (to be of basaltic composition), (2) domes and composite cones, (3) highland paterae, and related (4) volcano-tectonic features. Many plains units like Lunae Planum and Hesperia Planum are thought to be of volcanic origin, fed by morphologically clearly-defined volcanoes or by huge fissure volcanism. Many small volcanic cone fields in the northern plains are interpreted as cinder cones (Wood, 1979), formed by lava and ice interaction (Allen, 1979; Greeley and Fagents, 2001), or as the products of phreatic eruptions (Frey et al., 1979). An overview of the temporal distribution of processes, including the volcanic activity as well as the erosional processes manifested by large outflow channels ending in the northern lowlands and sculpting large units of the volcanic flood plains is given by Neukum and Hiller (1981). Plescia and Saunders (1979) and Neukum and Hiller (1981) both gave chronological classifications of

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Table 1 (continued)

Volcanic construct	Area	Cum. number (≥ 1 km)	Ages in Ga
Albor Tholus	Caldera	7.99e-4	1.6
Hecates Tholus #3			
Caldera	Caldera 1	4.97e-4	1.02
	Caldera 2	1.57e-4	0.322
	Caldera 3	1.36e-4	0.28
	Caldera 4	4.4e-5	0.09
	Caldera 5	5.36e-5	0.11
Side-Caldera and Flanks	Caldera ejecta	1.74e-4	0.357
	Flank dissected	4.52e-4	0.927
	Flank air fall	4.26e-4	0.86
Elysium plains		2.01e-4	0.411
Highland Volcanoes			
Apollinaris Patera	Caldera small	4.08e-3+	3.61+
	Caldera big	6.82e-3	3.72
	Shield	1.11e-2/7.58e-3+	3.81/3.74+
	Fan (south)	6.65e-3	3.71
Circum Hellas volcanoes			
Hadriaca Patera #4	Caldera	5.26E-4-3.22E-3	1.08-3.54
	Flank	5.62E-4-1.52E-2	1.15-3.86
Tyrrhena Patera	Caldera	1.85e-3/7.20e-4+	3.29/1.48+
	Flank	2.32e-2/5.24e-3+/1.43e-3+	3.93/3.66+/2.90+
	AhrF	1.79e-3/5.38e-4	3.26/1.10
	He	6.94e-4	1.42
	Nsu1+2	3.33e-2/3.04e-3+/7.68e-4+	3.98/3.53+/1.58+
	Ns1	3.98e-3/1.25e-3+	3.0/2.56+
	Ns2	1.96e-3/2.84e-3+/9.07e-4+	3.90/3.50+/1.86+
	AHcf1	1.59e-3/5.88e-4+	3.11/1.21+
	NsL1-4	2.52e-3/7.87e-4+	+ 3.46/1.61+
Amphitrites Patera	Caldera	7.69e-3/3.51e-3+	3.74/3.57+
Peneus Patera #5	Edifice	8.09e-3	3.75
Malea Patera #5	Edifice	1.11e-2	3.81
Pityusa Patera	Edifice	1.05e-2	3.80
Syrtis Major volcanoes			
Meroe Patera	Caldera	7.28e-3/1.13e-3+	3.73/2.33+
Nili Patera		No counts	-

+ Resurfacing ages are determined according to treatment description in Appendix B.

#1 Compare Basilevsky et al. (2006).

#2 Compare Basilevsky et al. (2005).

#3 Compare Neukum et al. (2004).

#4 Compare Zuschneid (2005) and Williams et al. (2007).

#5 Compare Williams et al. (2008b).

counts, absolute ages in partially resurfaced units and errors in the relative and absolute age determination.

2. The Tharsis volcanic province surface ages

The Tharsis region is the most dominating feature of the martian topography and shows numerous volcanic constructs of different age and morphology.

2.1. Alba Patera

The northernmost volcano of the Tharsis assemblage is Alba Patera (Fig. 1), which is a gigantic volcanic shield with a base diameter of roughly 1100 km, even larger than the enormous Olympus Mons. Compared to Olympus Mons, Alba Patera is a wide but low-relief construct about 6 km in height, with flank slopes of about 1° and calderas extending to 120 km in diameter. The surrounding flank grabens (Tantalus and Alba Fossae) extend in a North-South direction. There are extensive lava flows and local dendritic valleys found at the summit region around the caldera. The volcano is divided into two parts indicating at least two formation stages; one broad lower construct (about 4–5 km high) cut by the Fossae and marginal lava aprons (Ivanov and Head, 2003), and a much smaller summit shield (about 1 km in height) which contains a caldera and is situated on top of a broad summit plateau in the shal-

lower construct. Age determinations by Neukum and Hiller (1981) yielded surprisingly young ages. For comparison with the absolute ages described here, the collection of relative ages of Neukum and Hiller (1981) are recalculated on the basis of their crater retention ages N_{cum} ($D \geq 1$ km), applying the later chronology model of Hartmann and Neukum (2001) and shown in Fig. 2.

Based on Viking and HRSC data, Werner (2005) confirmed the maximum age of about 3.5 Ga by measurements at the northern flank base of Alba Patera, an area where the dichotomy scarp cannot be seen. Measurements based on the HRSC image taken during orbit 1272 at the lower flank (north of the summit) support this oldest age deduced from the low-resolution Viking image data. At the western flank of Alba Patera, I found ages indicating at least four episodes of activity at about 3.4 Ga, 2 Ga, 800 Ma ago, and as recent as 250 Ma ago (Fig. 3). All other crater counts have been made in the upper part of the construct and indicate similar ages of about 1.1 Ga to 3 Ga. Resurfacing is possibly related to the formation of sinuous channels, which probably eroded through flowing lava as seen on other martian volcanoes, such as Hecates Tholus (Williams et al., 2005). Fluvial erosion by snow-pack melting possibly related to volcanic activity (Fassett and Head, 2007) could also explain such channel features. The youngest ages are found in the closest vicinity of the large caldera, yielding two episodes that ended as recently as between 800 Ma and 180 Ma ago. This two-stage activity is supported by the summit caldera morphol-

2.3. Olympus mons

The largest and most prominent martian shield volcano is Olympus Mons, which has a relief of about 22 km and a basal

extent of about 800 km by 600 km measured from the edges of the scarp surrounding the main edifice. The width is almost doubled if the enormous extent of the enigmatic aureole deposit is included, which is dispersed over many hundreds of kilometers

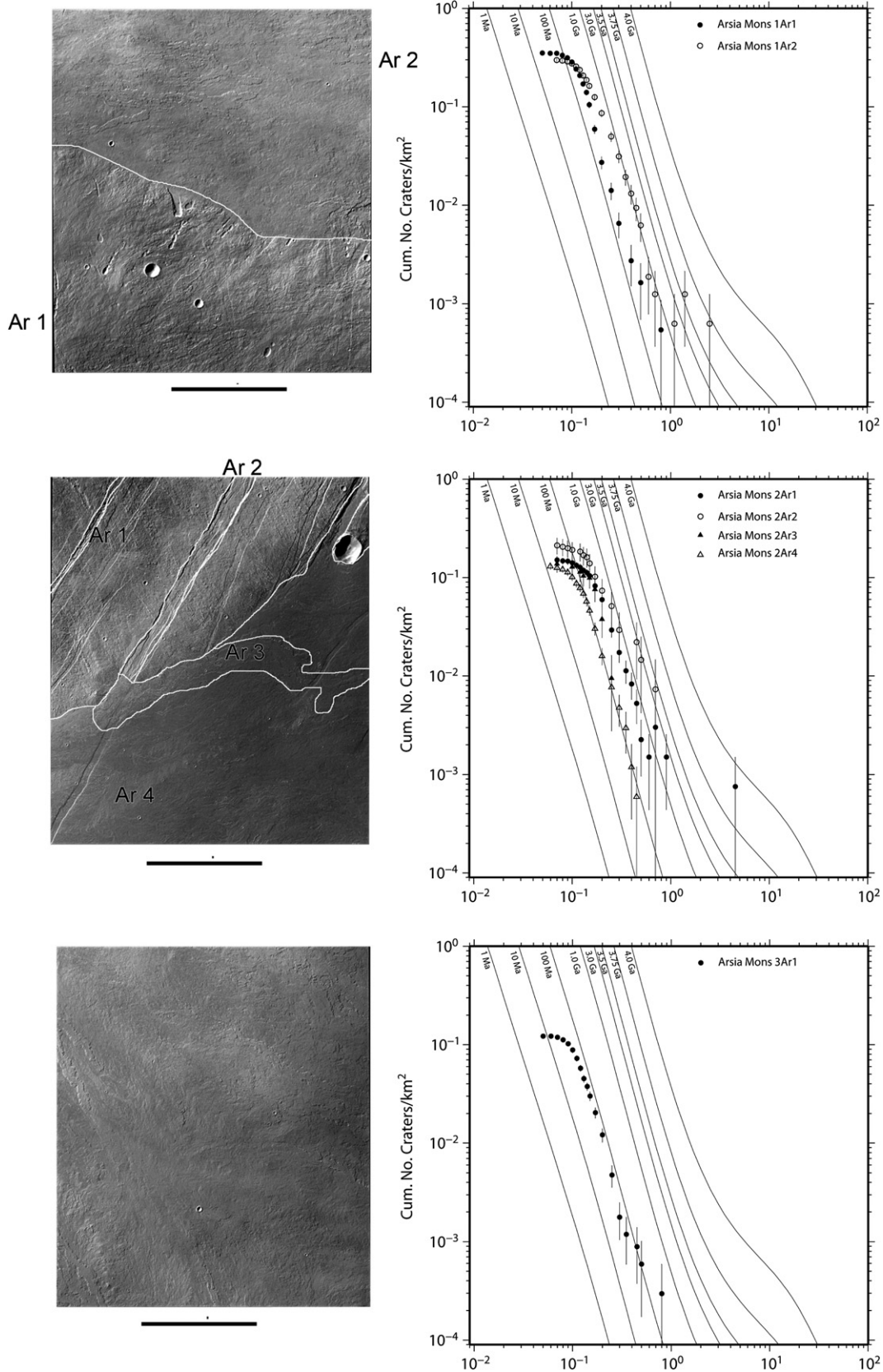


Fig. 4. The details of HRSC orbit 957 cover the flanks of Arsia Mons, on which units were counted. The scale indicates 10 km, if not said otherwise.

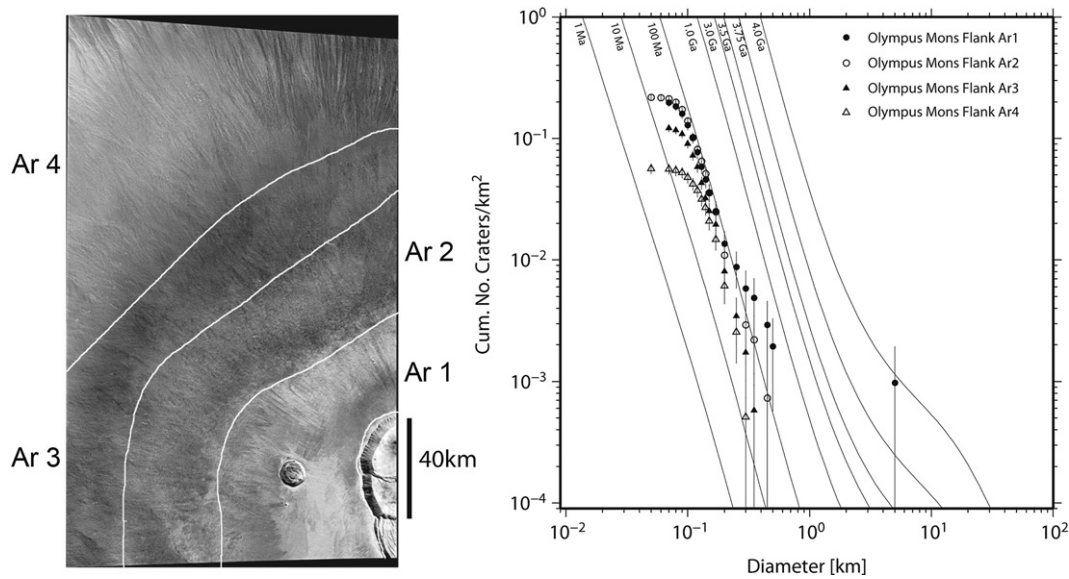


Fig. 8. The details of HRSC orbit 68 cover the flanks of Olympus Mons, on which units were counted.

uation constrains the sequence, while the ages derived from a set of crater size-frequency measurements yield individual fit ages of around 150 Ma, although they are statistically indistinguishable within the error limits (Fig. 7). The averaged surface age found for the Olympus Mons' caldera is in good agreement with earlier measurements (Hartmann, 1999; Hartmann and Neukum, 2001), which were based on MOC images in small areas of the calderas of Olympus Mons as well as Arsia Mons. This indicates that the summits of these edifices have been essentially active almost up to the present.

The vicinity of the caldera is characterized by superposing lava flows, representing the latest active phases. Measuring the flank surface ages in four units around the caldera, the average surface age is oldest close to the caldera (resurfacing ended about 210 Ma ago), and ages of about 170 Ma are found further down-slope and elsewhere on the flanks. Morphologically, the coverage by lava flows is more effective further down-slope than in the close vicinity of the caldera, which is also reflected in the crater size-frequency distribution. Here, apart from a few large impact craters that are embayed by younger lava flows (one 10.5 km crater is visible in Fig. 8), the flank surface is not completely covered by flows and thus part of an older 700 Ma-aged surface has survived. This agrees with widely-observed flank eruptions on Mars and Earth. Using the observed crater size-frequency distribution, the effect of geological resurfacing is visible in the shape of the distribution (Neukum and Horn, 1976). The crater diameter (and derived rim height) at which the crater size-frequency distribution is affected, yield a thickness of an uppermost layer a few hundred meters deposited within about 500 Ma.

The summit plateau is truncated by an up to 7 km high scarp, which is present in most places but is occasionally modified by lava covering and flattening of the steep flanks. Detailed investigation of the western scarp morphology has been based on HRSC and MOC imagery (Neukum et al., 2004; Basilevsky et al., 2005, 2006). The western flank edge exposes a ridge and several smaller mesas. The ages of the mesa surface suggest that they could be remnants of the very early and ancient **proto-Olympus Mons**. Ages of about 3.8 Ga were determined by measurements in the aureole (Hiller et al., 1982), supporting the notion that most of the volcanic construct was emplaced very early in martian history. Prominent and thin layering is visible at the steep slopes of the scarp, indicating the volcanic origin of the plateau, and Basilevsky et al. (2005) identified different slope types. One type (S2 slope)

is found only at the western flank of Olympus Mons, whilst at the upper part several chaos-like depressions occur from which channel-like grooves evolve down-slope. This morphology may relate to the influence of water (see Basilevsky et al., 2005). The ages derived from crater size-frequency measurements for the **volcanic units** on the flanks range between 500 Ma until very recently, indicating that the flank was blanketed by **several episodes at about 500 Ma, 200 Ma, and 100 Ma**. However, many of the measured crater size-frequency distributions show relatively flat distributions that indicate the steady (in terms of millions of years) supply of new lava flows, although they did not fully blanket the earliest crater record, thus putting the **essential erection phase of Olympus Mons in the earliest part of martian evolutionary history (before 3.6 Ga ago)**.

2.4. The Tholi and Paterae on Tharsis

Eight martian volcanoes are classified as shields: Olympus Mons, Ascreaus Mons, Pavonis Mons, Arsia Mons, Alba Patera, Biblis Patera, Uranus Patera, and Jovis Tholus, but a few dome-type volcanoes are also present in the Tharsis region. Northeast in the prolongation of the Tharsis Montes chain, the Uranus Group consists of Uranus Patera, Uranus Tholus, and Ceraunius Tholus. To the south and east of the Tharsis Montes lies Tharsis Tholus. These four volcanoes all have a basal diameter of between 100 and 300 km and a relief ranging from about 3 km (Uranus Patera and Uranus Tholus) to around 6 km (Ceraunius and Tharsis Tholi). Most have an asymmetric shape and a multi-stage caldera which occupies large portions of the visible construct. Many of their flanks are buried under lava flows of the surrounding plains, generated by the Tharsis Montes, so that their true dimensions remain unknown.

Uranus Patera appears to have a complex development, indicated by fan-shaped segments emanating from its complex caldera. New age determinations from low-resolution HRSC limb observation data yield a shield and caldera age of about 3.7 Ga, while a resurfacing event, which ended about 3.5 Ga ago, affected the caldera and parts of the flanks. Higher resolution HRSC images support the impression that even later resurfacing, unrelated to volcanic activity, has occurred (Fig. 9a). *Uranus Tholus* is a small cone-like volcano with a two-stage caldera, representing at least two active phases. New crater counts confirm that the construct had been built up at 4.04 Ga ago, while the caldera floor was em-

have been accompanied by age dating. Those results are included in the discussion presented here. The crater count results and derived relative and absolute ages of this study are listed in Table 1, and these along with published results (Basilevsky et al., 2005, 2006; Hauber et al., 2005; Neukum et al., 2004; Werner et al., 2003a, 2003b, 2004, 2005; Werner, 2005; Williams et al., 2007, 2008a) are plotted in Fig. 2. All resulting crater counts gathered in this work are presented together with measurements based on Viking image data (Fig. 2), which were assembled by Neukum and Hiller (1981), who evaluated measurements published by Blasius (1976); Carr (1976); Carr et al. (1977); Crumpler and Aubele (1978); Masursky et al. (1977); Neukum and Wise (1976); Plescia and Saunders (1979); Wise et al. (1979b) and their own.

The novelty of the global data set presented and interpreted here is that counts have been performed by a single observer, they are based on a single set of high-resolution image data, and a single applied crater production function and chronology model. This guarantees a very coherent set of data and allows for a better comparison of the results and subsequent comparison with earlier measurements. Based on the results shown in Fig. 2, the following volcanic evolution for Mars is derived.

All volcanic constructs were formed and built up to their present size early in martian history (before about 3.6 Ga ago). This implies that most of the volcanism had started about 4 Ga ago or even earlier. In that time period, the highland units had already been emplaced and inter-crater plains were formed. Possibly even the basement of the lowlands had formed about that time (before 3.8 Ga ago). The small tholi and paterae in the Tharsis region are relics of such an early volcanic period. In their case, the vent production of magma stopped about 4 Ga ago and shows younger activity until about 3.7 Ga ago only in a few cases. Few of the tholi and paterae show later resurfacing, which is not clearly related to volcanic processes. Global volcanic activity during the early martian history is manifested in many small shields or domes found for example in the vicinity of Apollinaris Patera or in the Coracis Fossae, where we obtained ages of about 3.9 Ga for the latter (Grott et al., 2005). A similar decrease in activity is found for the highland volcanoes: Meroe Patera (3.73 Ga; Syrtis Major), Apollinaris Patera (3.71 Ga), and Amphitrites Patera (3.74 Ga), with no obvious volcanic activity occurring later than 3.6 Ga ago. Subsequent resurfacing is observed and could be related to a blanketing deposition of unknown origin which is seen both inside and outside the calderas. A more varied timing of events is observed for Hadriaca and Tyrrhena Paterae, situated east of Hellas in Hesperia Planum, a volcanic plain that formed about 3.71 Ga ago. The evolutionary history of both volcanoes appears synchronized in processes and timing. The very shallow broad shields, composed of material from more explosive volcanic activity, were already constructed by about 3.9 Ga ago. Subsequent activity resurfaced the vicinity by possible ash deposition and by later carving channels into the unstable ash deposits until about 3.7 Ga ago. The eruption style later changed to a more effusive one as observed at their crest regions (about 3.5 to 3.3 Ga ago), and as indicated by the large caldera-flank channel on Tyrrhena Patera. Inside the Hadriaca caldera and at Tyrrhena Patera, the youngest construct-wide resurfacing event ended about 1.6 Ga ago. Ages reported by Williams et al. (2008b) for Malea, Peneus, and Pityusa Paterae indicate that they formed before 3.75 Ga ago. The Elysium volcanoes formed before 3.7 Ga ago and reached their final size about 3.5 Ga ago. Extensive volcanic activity is observed at the western flanks of the Elysium rise during a period between 3.4 and 3.3 Ga ago. Enormous volcanic flows expand into Utopia Planitia, while flank failures support the formation of channels by the release of water. The surrounding plains to the northeast (Arcadia Planitia) indicate subsequent volcanic flooding about 2.6 Ga ago, but the source remains unclear. The caldera and upper flanks of Elysium Mons

formed about 3.5 Ga ago, showing a resurfacing about 1.6 Ga that cannot be related to volcanic activity because both ages are found inside and outside the caldera for both Hadriaca and Tyrrhena Paterae. Measurements especially at Hecates Tholus but also at Albor Tholus indicate that volcanic activity occurred later than previously thought. Caldera and flank ages show subsequent resurfacing over the past 2 Ga, while at Hecates Tholus this happened over the past 1 Ga until about 100 Ma ago. Elysium Planitia, a region southeast of the Elysium rise, is one of the youngest plains on Mars (formed between 10 and 30 Ma ago). The ages measured in this region have been interpreted as a result of the discovery of crater Zunil and its numerous secondary-crater strewn field (McEwen et al., 2005). As discussed in Werner et al. (2009), the misinterpretation in age is less than a factor of two if secondary craters were included in the measurements unwittingly.

For the other large Tharsis volcanoes, crater size-frequency distributions indicate that they reached their final size about 3.55 Ga ago, with later more or less intensive resurfacing of the flanks. At all shield flanks, episodes of volcanic eruptions (flows) are observed, which have surface ages of between 500 Ma and 100 Ma. The youngest ages determined by the crater size-frequency measurements are about 2 Ma (Olympus Mons escarpment), suggesting that the volcanoes are potentially still active (Neukum et al., 2004). For Alba Patera, the most recent flank resurfacing occurred about 200 Ma ago, possibly as the result of volcanic activity. Caldera floor ages of the Tharsis Montes and Olympus Mons show that the latest vent activity happened between 200 Ma and 100 Ma ago. At the foot of Pavonis Mons, small shields are observed that formed about 300 Ma to 100 Ma ago. This latest extensive volcanic activity, observed in both large volcanic provinces (Tharsis and Elysium), correlates well with crystallization ages found for basaltic martian meteorites (Shergottites) and indicates that the applied chronology model accurately reflects the surface ages. However, recently Bouvier et al. (2005, 2008) argued that the young Shergottite ages represent hydrothermal alteration and the crystallization age is 4.1 Ga.

This study is the first to constrain the age for the formation of the Medusae Fossae at about 1.6 Ga. This correlates with late-stage resurfacing of some highland volcanoes (Hadriaca and Tyrrhena Paterae) and at least some ages found at the flank of the Tharsis Montes. The same age is observed for some volcanic surfaces where no clear source can be identified (Elysium Mons, Albor Tholus, Meroe Patera). If the Medusae Fossae deposits are interpreted as pyroclastic ash that is widespread over the planet and accumulated at the dichotomy boundary, it is possible that this material could have been deposited as a thin layer in many places, for example, at the volcanoes. This possible final stage of global volcanic activity is perhaps supported by the age of another group of martian meteorites (Nakhlites). Crater size-frequency measurements confirm that most edifices were constructed over billions of years and are characterized by episodic phases of activity that continued in both large volcanic regions almost to the present. A number of caldera floor and flank ages cluster around 150 Ma, indicating a relatively recent peak activity period that practically coincide with radiometrically measured ages of Shergottites. The relation to the second group of martian meteorites, the Nakhlites, remains more speculative. Most of the smaller volcanoes in the Tharsis region were active in early martian geological history, similar to most of the highland volcanoes. The long activity of martian volcanoes correspondingly implies a long lifetime of the “feeding” source especially in the Tharsis region that indicates a long and stable dynamic regime in the planet’s interior.

8. Implications

The volcanic and related tectonic activity sheds light on the internal dynamics and thermal evolution of planets. Essential time-

markers are set by understanding the evolutionary history of martian volcanic constructs. After providing a time-frame for surface processes, the timing for the thermodynamical evolution of Mars can now be assessed. The apparent absence of plate tectonics on Mars and the presence of magnetized crustal materials, requiring the presence of a strong ancient dynamo field (Acuña et al., 1999), makes Mars an interesting planet to compare with Earth or other inner Solar System bodies. Many aspects of the scenario of the early and subsequent evolution, including internal dynamics of Mars, have been discussed earlier (most recently by Spohn et al., 2001; Solomon et al., 2005; Nimmo and Tanaka, 2005). Usually, the timing is based on educated guesses, while the results presented here essentially push the time-frame forward. In the early martian history, it is evident that volcanism occurred planet-wide and all volcanic regions were emplaced before about 3.5 Ga ago. It is also demonstrated that the Elysium and Tharsis regions experienced volcanic activity until very recently (i.e. 200 to 100 Ma ago, even less in a few places), suggesting that the most recent activities (over the last 500 Ma of martian history) are more widespread than previously thought.

Hand in hand with the understanding of the time frame of the decay of the martian magnetic field (Werner, 2008), the thermal evolution and the internal dynamics are more constrained. While large-scale convection in the mantle is responsible for the continuous reshaping of Earth's surface through plate tectonics and volcanism, plate tectonics is currently not observed on Mars. The absence of plate tectonics and lower martian gravity could be a reason for the larger size of the martian volcanoes (BVSP, 1981).

On Mars, volcanism is extensive but not uniformly distributed, and includes a diversity of volcanic landforms. The Tharsis and the Elysium regions are two volcanic provinces that are topographically dominating (Fig. 1), situated close to the equator on the dichotomy boundary between the cratered (older) highlands and the northern lowlands (about 120° apart). A possible correlation of young volcanism and areoid (martian geoid) highs could suggest mantle upwelling (Steinberger et al., 2009). The regions where upwelling might be indicated in the areoid are characterized by volcanoes (Olympus Mons, Ascraeus Mons, Pavonis Mons and Arsia Mons) whose morphologies are strongly analogous to volcanic landforms on Earth.

In order to explain the internal dynamics of planets, especially that of Mars, the following scenario, summarized by Spohn et al. (2001) is required: After accretion, a planet differentiates into a core, mantle, crust and atmosphere/hydrosphere of unknown conditions. Several accretion and differentiation models have been postulated and usually incorporate time scales in the order of 20 to 50 Ma. Mars's mass and moment-of-inertia factor today supports a differentiated structure. The subsequent thermal evolution or cooling history is transferring heat by conduction and by thermal convection to remove the heat generated by radioactive nuclides and to cool the interior. Large-scale convection has been considered to form a crustal dichotomy as derived from Mars Global Surveyor gravity and topography data (Zuber et al., 2000). Numerical models show, simultaneously to a cooling core, that mantle flow is dominated by widely distributed down-welling. Broad local upwelling flows are commonly found in internally heated convection models (Spohn et al., 2001 and references therein). In the case of Mars, a hotter (or superheated) core probably allowed for a strong dynamo in the early history of Mars. A spinel-perovskite phase transition, even though Mars's temperature and pressure schemes are not well-defined, might have amplified large-scale and localization of upwelling flows. A cooling core reduces the plume activity with time. These scenarios are discussed by Spohn et al. (2001) who concluded that models that attempt to explain the martian evolution are still limited by many aspects such as dynamo theory, rheology, and mantle-lithosphere interaction. Recent volcanism might

no longer be driven by up-welling mantle convection (plumes) but due to crustal thickness growth and buoyancy (e.g., Schumacher and Breuer, 2007). Both ideas compete but no final proof for one or the other case can be given. In this sense, Earth appears unique among the terrestrial planets in possessing plate tectonics. Possibly, its mantle convection regime produces convective stresses to generate failure in the rigid surface boundary layer. Other planets (e.g., Venus) appear to be in a stagnant-lid regime; a regime which is characterized by the formation of a nearly immobile lid on top of a convective mantle that occurred due to large viscosity of the upper thermal boundary layer.

For Mars, the following sequence of events is suggested: Major (voluminous) volcanic activity occurred globally. The cessation of the magnetic dynamo (before 3.9 Ga ago) was followed by possible plume activity in all volcanic centres (Tharsis, Elysium, circum-Hellas and Syrtis Major region). The last observed global volcanic activity ended around 3.6 Ga ago. Plains-forming activity ended even earlier (3.7 Ga). Locally, more recent activity is observed in Elysium, Tharsis and possibly at Hadriaca and Tyrrhena Paterae. Models suggest that the pattern of mantle convection changed to a dominantly broad weak upwelling after the core cooled and the dynamo ceased as a consequence of too-low heat flux across the core-mantle boundary. Three broad areas of possible upwelling can be identified in the areoid (Steinberger et al., 2009). Volcanism and magmatism could continue as on Venus, localized in places where thick insulating crust formed (Schumacher and Breuer, 2007).

A comparison of dynamical patterns reflected on the planet's surface and its gravity anomalies and geoid patterns allow for a better understanding of the thermal evolution of planets. Work presented here strongly support that Mars is a planet which takes its place between the two end-members of terrestrial planetary evolution: While Mercury and the Moon appear to record the early history of the Solar System, and show no signs of water relevant to their evolution, Mars at least shows a very diverse surface history. It is a planet whose surface features show the total record of both internal and external processes from the beginning of its existence 4.5 Ga ago until now. Similarly, Barlow (1988) concluded from crater size-frequency distribution analysis on the basis of Viking image data, that Mars is the only place in the solar system to display surfaces from throughout the planet's history. The role of water in Mars's early history is unclear, but evidence for surface action of water is prominent (Carr, 1996). Venus and Earth are two end-members that show very young surfaces too. The major difference is the presence of water. While Venus is presumably dry, the hydrosphere on Earth might be key to plate tectonics. The validity of theories concerning (deep) plume activity and plate tectonics versus stagnant-lid and crustal thickening could gain additional support from the comparison between planets. Ultimately, these results will contribute to the knowledge of similarities and differences between planets, and thus improve our understanding on how each one of them evolves thermally in its interior and geologically on its surface.

Acknowledgments

The author thanks Nadine Barlow and Doris Breuer for their comments, discussion and help to improve the manuscript. Acknowledgments to Gerhard Neukum for the opportunity to use the HRSC data in first hand and provided funds from the Deutsche Forschungsgemeinschaft (DFG). The author appreciated the discussion within the HRSC science team, which led to many publications included in this paper, and acknowledges the use of Mars Orbiter Camera and THEMIS image data, as well thanks the HRSC Experiment Teams at DLR Berlin and the Freie Universität Berlin, and the Mars Express Project Teams at ESTEC and ESOC for their successful planning and acquisition of data.